# Polar Form, De Moivre's Theorem & Roots of Unity

## Section 1: Polar and Euler Forms

**1** Express $z = -2 - 2\sqrt{3}i$ in Euler's exponential form $re^{i\theta}$ with $\theta \in (-\pi, \pi]$.

**a.** $4e^{-i2\pi/3}$

**b.** $4e^{i4\pi/3}$

**c.** $2e^{-i2\pi/3}$

**d.** $4e^{-i\pi/3}$

**2** Convert $z = 6e^{i3\pi/4}$ into exact Cartesian form $x + iy$.

**a.** $-3\sqrt{2} + 3\sqrt{2}i$

**b.** $3\sqrt{2} - 3\sqrt{2}i$

**c.** $-3 + 3\sqrt{3}i$

**d.** $-3\sqrt{2} - 3\sqrt{2}i$

**3** If $z_1 = 3e^{i\pi/3}$ and $z_2 = 2e^{i\pi/6}$, evaluate the quotient $\frac{z_1}{z_2}$ in exponential form.

**a.** $\frac{3}{2}e^{i\pi/6}$

**b.** $\frac{3}{2}e^{i\pi/2}$

**c.** $6e^{i\pi/6}$

**d.** $1.5e^{-i\pi/6}$

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## Section 2: De Moivre's Theorem for Powers

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**4** Use De Moivre's Theorem to evaluate $(1 + i\sqrt{3})^6$.

**a.** $64$

**b.** $-64$

**c.** $64i$

**d.** $-64i$

**5** Evaluate $(\cos(\pi/5) + i\sin(\pi/5))^{10}$.

**a.** $1$

**b.** $-1$

**c.** $i$

**d.** $-i$

**6** Express $\cos(3\theta)$ in terms of $\cos\theta$ using De Moivre's Theorem $(\cos\theta + i\sin\theta)^3$.

**a.** $4\cos^3\theta - 3\cos\theta$

**b.** $3\cos^3\theta - 4\cos\theta$

**c.** $\cos^3\theta - 3\cos\theta$

**d.** $4\cos^3\theta + 3\cos\theta$

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## Section 3: Roots of Complex Numbers

**7** Find all cube roots of $8i$ in polar form.

**a.** $2e^{i\pi/6}, 2e^{i5\pi/6}, 2e^{-i\pi/2}$

**b.** $2e^{i\pi/3}, 2e^{i\pi}, 2e^{i5\pi/3}$

**c.** $8e^{i\pi/6}, 8e^{i5\pi/6}, 8e^{-i\pi/2}$

**d.** $2,\ -1 + i\sqrt{3},\ -1 - i\sqrt{3}$

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**8** How are the $n$ distinct $n$th roots of any complex number $z = re^{i\theta}$ distributed on the Argand plane?

**a.** They lie on a circle of radius $r^{1/n}$ centered at the origin, evenly spaced by $\frac{2\pi}{n}$ radians

**b.** They lie along a straight line through the origin

**c.** They form an ellipse with eccentricity $1/n$

**d.** They cluster exclusively in Quadrant I

**9** What is the sum of all $n$ distinct $n$th roots of any complex number $z$ ($n \geq 2$)?

**a.** $0$

**b.** $1$

**c.** $n$

**d.** $z$

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## Section 4: Roots of Unity & Factorization

**10** If $\omega = e^{i2\pi/3}$ is a primitive complex cube root of unity, what is the value of $1 + \omega + \omega^2$?

**a.** $0$

**b.** $1$

**c.** $-1$

**d.** $3$

**11** Factor $z^4 + 1$ into two real quadratic factors.

**a.** $(z^2 - \sqrt{2}z + 1)(z^2 + \sqrt{2}z + 1)$

**b.** $(z^2 + 1)^2$

**c.** $(z^2 - 1)(z^2 + 1)$

**d.** $(z^2 - 2z + 1)(z^2 + 2z + 1)$

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**12** The four 4th roots of unity ($z^4 = 1$) form the vertices of which geometric shape on the Argand plane?

**a.** A square inscribed in the unit circle with vertices at $\pm 1, \pm i$

**b.** An equilateral triangle

**c.** A regular hexagon

**d.** A rectangle of perimeter 4

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